

Retail Demand Forecasting Pipeline

Multi-Horizon Store-Level Sales Forecasting on the Rossmann Store Sales Dataset

1. Objective

The objective of this project is to build a leakage-safe, multi-horizon demand forecasting pipeline for retail store-level sales, and to quantitatively evaluate whether machine learning models (LightGBM, CatBoost) outperform a naive statistical baseline. The pipeline is designed end-to-end: from raw data acquisition, through feature engineering, model training, to rigorous time-series evaluation across multiple forecast horizons (7, 14, and 28 days).

2. Dataset

The project uses the Kaggle **Rossmann Store Sales** competition dataset, consisting of daily sales records for 1,115 German drugstore chains between January 1, 2013 and July 31, 2015.

Property	Value
Total records	1,017,209 rows
Number of stores	1,115
Date range	2013-01-01 to 2015-07-31
Core fields	Store, Date, Sales, Customers, Open, Promo, StateHoliday, SchoolHoliday
Store metadata	StoreType, Assortment, CompetitionDistance, Promo2, Promo2SinceWeek/Year

3. Methodology

3.1. Feature Engineering

The following feature groups were engineered, all computed strictly from past information to prevent lookahead leakage:

- Calendar features: day-of-week, month, week-of-year, quarter, weekend flag.
- Lag features: sales value 1, 7, 14, and 28 days prior to the target date.
- Rolling statistics: 7/14/28-day rolling mean and standard deviation, computed on data shifted by one day (`shift(1)`) so no feature uses same-day or future information.
- Store metadata: encoded StoreType and Assortment categories; CompetitionDistance with missing values imputed using the median.
- Exogenous regressors: Promo flag, combined Holiday flag (StateHoliday + SchoolHoliday), and store Open/Closed status.

3.2. Validation Strategy

A strict time-based (expanding-window style) train/test split was used: the final 42 days of the dataset were held out as the test set, with no random shuffling. This ensures the model is never trained on data from the future relative to what it is being evaluated on – the most common and critical pitfall in time-series forecasting projects.

3.3. Models Compared

- **Naive Baseline (lag-7)**: predicts sales using the value from exactly 7 days prior, capturing basic weekly seasonality without any learning.

- **LightGBM:** gradient-boosted decision tree regressor with native categorical handling for the store identifier; 300 boosting rounds, learning rate 0.05, 64 leaves.
- **CatBoost:** gradient-boosted decision tree regressor with native categorical handling for the store identifier (no manual one-hot encoding); 200 iterations, learning rate 0.08, depth 8.

3.4. Evaluation Metrics

Models were evaluated at three forecast horizons – 7, 14, and 28 days – using three standard forecasting metrics: RMSE (Root Mean Squared Error), MAE (Mean Absolute Error), and WAPE (Weighted Absolute Percentage Error). Evaluation was restricted to days when the store was open, since closed-store days have zero sales by definition and would artificially inflate accuracy.

4. Results

The table below reports actual results obtained from running the pipeline end-to-end on the Rossmann dataset.

Horizon (days)	Model	RMSE	MAE	WAPE (%)
7	Naive (lag-7)	2879.08	2271.23	37.74
7	LightGBM	758.64	474.92	7.89
7	CatBoost	767.26	473.53	7.87
14	Naive (lag-7)	3207.34	2483.01	35.24
14	LightGBM	990.93	633.90	9.00
14	CatBoost	1021.63	640.34	9.09
28	Naive (lag-7)	2974.26	2277.99	32.51
28	LightGBM	897.39	593.00	8.46
28	CatBoost	934.39	613.43	8.75

Highlighted rows indicate the best-performing model (lowest WAPE) at each horizon.

4.1. Key Findings

- Both LightGBM and CatBoost reduce WAPE from 32–38% (naive baseline) to 8–9% across all three horizons – a 74–79% reduction in forecast error.
- At the 7-day horizon, CatBoost marginally outperforms LightGBM (WAPE 7.87% vs 7.89%).
- At the 14 and 28-day horizons, LightGBM performs marginally better (WAPE 9.00% and 8.46% respectively) than CatBoost.
- Forecast error remains stable (8–9% WAPE) as the horizon extends from 7 to 28 days, indicating the lag and rolling features generalize well beyond a single-step forecast.
- The large gap between the naive baseline and both ML models confirms that the engineered features carry genuine predictive signal rather than the models simply memorizing recent values.

5. Data Leakage Prevention

Time-series forecasting projects are highly prone to inflated, misleading accuracy caused by data leakage. The following safeguards were explicitly implemented in this pipeline:

- All lag features use values strictly from prior time steps – never the current or future day.
- All rolling mean/std features are computed on data shifted by one day (`shift(1)`) before the rolling window is applied, ensuring the current day’s value never contributes to its own feature.
- The train/test split is purely chronological (no random shuffling), so the model is never trained on data that occurs after the test period begins.

- Evaluation excludes days when stores were closed, avoiding artificial accuracy inflation from trivial zero-sales predictions.

6. Reproducibility

The entire pipeline is contained in a single script, `retail_demand_forecasting.py`. Running the script performs data acquisition, feature engineering, model training, and evaluation in one pass.

Command to run

```
pip install pandas numpy lightgbm catboost scikit-learn requests
python retail_demand_forecasting.py
```

Output artifacts produced

- `train.csv`, `store.csv` – raw Rossmann data.
- `rossmann_features.csv` – feature-engineered dataset used for training and evaluation.
- `lightgbm_model.txt` – trained LightGBM model file.
- `catboost_model.cbm` – trained CatBoost model file.
- `model_comparison_results.csv` – final RMSE / MAE / WAPE comparison table.

7. Conclusion

This project demonstrates a complete, reproducible, leakage-safe retail demand forecasting pipeline validated on a large-scale dataset (1M+ records, 1,115 stores). Both LightGBM and CatBoost achieve an approximately 74–79% reduction in WAPE relative to a naive lag-based baseline, with stable performance across 7, 14, and 28-day forecast horizons. The results confirm that gradient-boosted tree models, combined with carefully engineered lag, rolling, and calendar features, provide a strong and practical approach to multi-horizon retail demand forecasting.